

CMP5367: Artificial Intelligence and Machine Learning

Lectures 8: Feature Engineering

Instructor: **Dr. Hadeel Saadany**

November 10th 2025

Outline

- Principal Component Analysis
- Handling Outliers
- Unbalanced Data
- Interpreting results with unbalanced data

What is Feature Engineering?

Feature engineering is the process of **creating, transforming, or selecting input variables** to improve a machine learning model's performance. It involves:

- Extracting meaningful patterns from raw data
- Encoding, scaling, or combining features.
- Making data more predictive and model-friendly
- Reducing dimensionality to remove redundancy or noise.
- In short: **turning raw data into useful signals for learning.**

Feature Selection

Drain(1)	DrainV(1)	Gate(1)	GateV(1)	Drain(2)	DrainV(2)	Gate(2)	GateV(2)	Drain(3)	DrainV(3)	Gate(3)	GateV(3)	Drain(4)	DrainV(4)	Gate(4)	GateV(4)	Drain(5)	DrainV(5)	Gate(5)	GateV(5)
3.29E-07	0.00E+00	1.52E-07	1.00E+00	2.97E-07	0.00E+00	1.57E-07	9.00E-01	2.57E-07	0.00E+00	1.83E-07	8.00E-01	2.19E-07	0.00E+00	1.62E-07	7.00E-01	1.83E-07	0.00E+00	1.42E-07	6.00E-01
4.33E-06	1.00E-02	1.93E-07	1.00E+00	4.24E-06	1.00E-02	1.77E-07	9.00E-01	4.14E-06	1.00E-02	1.72E-07	8.00E-01	4.01E-06	1.00E-02	1.88E-07	7.00E-01	3.87E-06	1.00E-02	1.52E-07	6.00E-01
1.27E-05	2.00E-02	1.47E-07	1.00E+00	1.22E-05	2.00E-02	1.22E-07	9.00E-01	1.17E-05	2.00E-02	1.52E-07	8.00E-01	1.11E-05	2.00E-02	1.88E-07	7.00E-01	1.05E-05	2.00E-02	2.18E-07	6.00E-01
1.91E-05	3.00E-02	1.42E-07	1.00E+00	1.83E-05	3.00E-02	1.62E-07	9.00E-01	1.75E-05	3.00E-02	1.62E-07	8.00E-01	1.66E-05	3.00E-02	1.47E-07	7.00E-01	1.56E-05	3.00E-02	1.62E-07	6.00E-01
2.55E-05	4.00E-02	1.32E-07	1.00E+00	2.44E-05	4.00E-02	8.11E-08	9.00E-01	2.33E-05	4.00E-02	2.38E-07	8.00E-01	2.21E-05	4.00E-02	1.62E-07	7.00E-01	2.08E-05	4.00E-02	1.27E-07	6.00E-01
3.18E-05	5.00E-02	1.01E-07	1.00E+00	3.05E-05	5.00E-02	1.27E-07	9.00E-01	2.91E-05	5.00E-02	1.52E-07	8.00E-01	2.76E-05	5.00E-02	1.22E-07	7.00E-01	2.60E-05	5.00E-02	1.98E-07	6.00E-01
3.81E-05	6.00E-02	1.01E-07	1.00E+00	3.66E-05	6.00E-02	1.17E-07	9.00E-01	3.49E-05	6.00E-02	1.62E-07	8.00E-01	3.31E-05	6.00E-02	1.52E-07	7.00E-01	3.11E-05	6.00E-02	1.52E-07	6.00E-01
4.45E-05	7.00E-02	6.08E-08	1.00E+00	4.26E-05	7.00E-02	1.72E-07	9.00E-01	4.07E-05	7.00E-02	1.62E-07	8.00E-01	3.85E-05	7.00E-02	1.06E-07	7.00E-01	3.62E-05	7.00E-02	1.17E-07	6.00E-01
5.07E-05	8.00E-02	1.52E-07	1.00E+00	4.86E-05	8.00E-02	1.17E-07	9.00E-01	4.64E-05	8.00E-02	1.52E-07	8.00E-01	4.39E-05	8.00E-02	1.98E-07	7.00E-01	4.13E-05	8.00E-02	1.57E-07	6.00E-01
5.71E-05	9.00E-02	1.62E-07	1.00E+00	5.47E-05	9.00E-02	1.22E-07	9.00E-01	5.21E-05	9.00E-02	1.62E-07	8.00E-01	4.94E-05	9.00E-02	1.88E-07	7.00E-01	4.63E-05	9.00E-02	1.67E-07	6.00E-01
6.34E-05	1.00E-01	7.61E-08	1.00E+00	6.07E-05	1.00E-01	1.72E-07	9.00E-01	5.79E-05	1.00E-01	1.93E-07	8.00E-01	5.47E-05	1.00E-01	2.03E-07	7.00E-01	5.14E-05	1.00E-01	1.62E-07	6.00E-01
6.97E-05	1.10E-01	1.57E-07	1.00E+00	6.68E-05	1.10E-01	1.47E-07	9.00E-01	6.36E-05	1.10E-01	1.42E-07	8.00E-01	6.01E-05	1.10E-01	2.43E-07	7.00E-01	5.65E-05	1.10E-01	1.27E-07	6.00E-01
7.60E-05	1.20E-01	1.27E-07	1.00E+00	7.29E-05	1.20E-01	1.47E-07	9.00E-01	6.93E-05	1.20E-01	1.37E-07	8.00E-01	6.56E-05	1.20E-01	1.77E-07	7.00E-01	6.15E-05	1.20E-01	1.32E-07	6.00E-01
8.24E-05	1.30E-01	1.06E-07	1.00E+00	7.88E-05	1.30E-01	1.32E-07	9.00E-01	7.51E-05	1.30E-01	1.47E-07	8.00E-01	7.09E-05	1.30E-01	1.83E-07	7.00E-01	6.64E-05	1.30E-01	1.57E-07	6.00E-01
8.86E-05	1.40E-01	1.12E-07	1.00E+00	8.48E-05	1.40E-01	1.42E-07	9.00E-01	8.07E-05	1.40E-01	1.42E-07	8.00E-01	7.62E-05	1.40E-01	1.67E-07	7.00E-01	7.15E-05	1.40E-01	1.12E-07	6.00E-01
9.50E-05	1.50E-01	1.06E-07	1.00E+00	9.09E-05	1.50E-01	1.83E-07	9.00E-01	8.64E-05	1.50E-01	1.62E-07	8.00E-01	8.16E-05	1.50E-01	2.38E-07	7.00E-01	7.64E-05	1.50E-01	1.47E-07	6.00E-01
1.01E-04	1.60E-01	1.01E-07	1.00E+00	9.68E-05	1.60E-01	1.93E-07	9.00E-01	9.21E-05	1.60E-01	2.18E-07	8.00E-01	8.69E-05	1.60E-01	1.06E-07	7.00E-01	8.13E-05	1.60E-01	1.32E-07	6.00E-01
1.08E-04	1.70E-01	8.62E-08	1.00E+00	1.03E-04	1.70E-01	1.32E-07	9.00E-01	9.78E-05	1.70E-01	1.32E-07	8.00E-01	9.22E-05	1.70E-01	1.67E-07	7.00E-01	8.64E-05	1.70E-01	1.88E-07	6.00E-01
1.14E-04	1.80E-01	6.08E-08	1.00E+00	1.09E-04	1.80E-01	1.42E-07	9.00E-01	1.03E-04	1.80E-01	1.72E-07	8.00E-01	9.76E-05	1.80E-01	1.83E-07	7.00E-01	9.13E-05	1.80E-01	1.93E-07	6.00E-01
1.20E-04	1.90E-01	1.22E-07	1.00E+00	1.15E-04	1.90E-01	1.22E-07	9.00E-01	1.09E-04	1.90E-01	1.57E-07	8.00E-01	1.03E-04	1.90E-01	2.33E-07	7.00E-01	9.61E-05	1.90E-01	1.57E-07	6.00E-01
1.27E-04	2.00E-01	1.32E-07	1.00E+00	1.21E-04	2.00E-01	1.32E-07	9.00E-01	1.15E-04	2.00E-01	1.12E-07	8.00E-01	1.08E-04	2.00E-01	2.43E-07	7.00E-01	1.01E-04	2.00E-01	1.47E-07	6.00E-01
1.33E-04	2.10E-01	1.77E-07	1.00E+00	1.27E-04	2.10E-01	1.32E-07	9.00E-01	1.20E-04	2.10E-01	1.32E-07	8.00E-01	1.13E-04	2.10E-01	1.57E-07	7.00E-01	1.06E-04	2.10E-01	1.83E-07	6.00E-01
1.39E-04	2.20E-01	1.42E-07	1.00E+00	1.33E-04	2.20E-01	1.37E-07	9.00E-01	1.26E-04	2.20E-01	1.88E-07	8.00E-01	1.19E-04	2.20E-01	1.88E-07	7.00E-01	1.11E-04	2.20E-01	1.52E-07	6.00E-01
1.45E-04	2.30E-01	1.32E-07	1.00E+00	1.39E-04	2.30E-01	1.88E-07	9.00E-01	1.32E-04	2.30E-01	1.62E-07	8.00E-01	1.24E-04	2.30E-01	1.52E-07	7.00E-01	1.16E-04	2.30E-01	1.57E-07	6.00E-01
1.52E-04	2.40E-01	1.22E-07	1.00E+00	1.45E-04	2.40E-01	1.88E-07	9.00E-01	1.37E-04	2.40E-01	1.77E-07	8.00E-01	1.29E-04	2.40E-01	1.32E-07	7.00E-01	1.20E-04	2.40E-01	1.42E-07	6.00E-01
1.58E-04	2.50E-01	8.62E-08	1.00E+00	1.51E-04	2.50E-01	2.03E-07	9.00E-01	1.43E-04	2.50E-01	1.47E-07	8.00E-01	1.34E-04	2.50E-01	1.62E-07	7.00E-01	1.25E-04	2.50E-01	1.27E-07	6.00E-01
1.64E-04	2.60E-01	1.12E-07	1.00E+00	1.57E-04	2.60E-01	1.77E-07	9.00E-01	1.49E-04	2.60E-01	1.47E-07	8.00E-01	1.39E-04	2.60E-01	1.72E-07	7.00E-01	1.30E-04	2.60E-01	1.12E-07	6.00E-01
1.71E-04	2.70E-01	9.13E-08	1.00E+00	1.63E-04	2.70E-01	1.52E-07	9.00E-01	1.54E-04	2.70E-01	1.57E-07	8.00E-01	1.45E-04	2.70E-01	1.12E-07	7.00E-01	1.35E-04	2.70E-01	1.67E-07	6.00E-01
1.77E-04	2.80E-01	1.01E-07	1.00E+00	1.69E-04	2.80E-01	1.42E-07	9.00E-01	1.60E-04	2.80E-01	1.52E-07	8.00E-01	1.50E-04	2.80E-01	1.06E-07	7.00E-01	1.39E-04	2.80E-01	1.52E-07	6.00E-01
1.83E-04	2.90E-01	1.17E-07	1.00E+00	1.75E-04	2.90E-01	1.12E-07	9.00E-01	1.65E-04	2.90E-01	1.37E-07	8.00E-01	1.55E-04	2.90E-01	9.63E-08	7.00E-01	1.44E-04	2.90E-01	1.77E-07	6.00E-01
1.90E-04	3.00E-01	6.59E-08	1.00E+00	1.81E-04	3.00E-01	1.57E-07	9.00E-01	1.71E-04	3.00E-01	1.72E-07	8.00E-01	1.60E-04	3.00E-01	1.17E-07	7.00E-01	1.49E-04	3.00E-01	1.57E-07	6.00E-01
1.96E-04	3.10E-01	6.08E-08	1.00E+00	1.87E-04	3.10E-01	1.32E-07	9.00E-01	1.76E-04	3.10E-01	1.27E-07	8.00E-01	1.65E-04	3.10E-01	1.06E-07	7.00E-01	1.54E-04	3.10E-01	1.22E-07	6.00E-01
2.02E-04	3.20E-01	1.01E-07	1.00E+00	1.92E-04	3.20E-01	1.12E-07	9.00E-01	1.82E-04	3.20E-01	1.06E-07	8.00E-01	1.70E-04	3.20E-01	8.11E-08	7.00E-01	1.58E-04	3.20E-01	1.37E-07	6.00E-01
2.09E-04	3.30E-01	1.01E-07	1.00E+00	1.99E-04	3.30E-01	1.22E-07	9.00E-01	1.87E-04	3.30E-01	1.27E-07	8.00E-01	1.75E-04	3.30E-01	1.12E-07	7.00E-01	1.63E-04	3.30E-01	1.47E-07	6.00E-01
2.15E-04	3.40E-01	1.22E-07	1.00E+00	2.04E-04	3.40E-01	2.18E-07	9.00E-01	1.93E-04	3.40E-01	9.63E-08	8.00E-01	1.81E-04	3.40E-01	1.52E-07	7.00E-01	1.67E-04	3.40E-01	1.67E-07	6.00E-01
2.21E-04	3.50E-01	9.63E-08	1.00E+00	2.10E-04	3.50E-01	1.77E-07	9.00E-01	1.99E-04	3.50E-01	1.32E-07	8.00E-01	1.85E-04	3.50E-01	1.27E-07	7.00E-01	1.72E-04	3.50E-01	1.77E-07	6.00E-01
2.27E-04	3.60E-01	1.17E-07	1.00E+00	2.16E-04	3.60E-01	1.47E-07	9.00E-01	2.04E-04	3.60E-01	1.77E-07	8.00E-01	1.90E-04	3.60E-01	1.57E-07	7.00E-01	1.77E-04	3.60E-01	1.52E-07	6.00E-01
2.34E-04	3.70E-01	1.06E-07	1.00E+00	2.22E-04	3.70E-01	2.48E-07	9.00E-01	2.09E-04	3.70E-01	1.88E-07	8.00E-01	1.96E-04	3.70E-01	1.42E-07	7.00E-01	1.81E-04	3.70E-01	2.08E-07	6.00E-01
2.40E-04	3.80E-01	9.63E-08	1.00E+00	2.28E-04	3.80E-01	1.32E-07	9.00E-01	2.15E-04	3.80E-01	1.22E-07	8.00E-01	2.01E-04	3.80E-01	9.63E-08	7.00E-01	1.86E-04	3.80E-01	1.77E-07	6.00E-01

You have a large dataset, and you want to use it to build a classifier, so that you can classify something (e.g. likelihood of network intrusion).

The data has 150 features

You need to cut it down to 15 features before you try machine learning. But the question is which 15?

Feature Selection: Reasons

- Select the most informative features for model training.
- Removing irrelevant data/ noise.
- Increasing predictive accuracy of learned models.
- Reducing the cost of the data.
- Improving learning efficiency, such as reducing storage requirements and computational cost.
- Reducing the complexity of the resulting model description, improving the understanding of the data and the model.

Feature Selection

The process of reducing the input features for the model by using only the relevant features and getting rid of noise in the data set is known as feature selection.



Student Name	Student ID	Pervious Marks	IQ	Attendance	Pass/Fail
Newton	123	80	90	High	Pass
Leonardo da vinci	456	89	95	High	Pass
Elon Musk	245	21	43	Low	Fail
Mark Zuckerberg	344	19	38	Mid	Fail

Feature Selection Methods

- Filter Methods

- Evaluate features based on **statistical measures or scores**.
- Features are selected on the basis of their **relevance scores** in various statistical tests for their correlation with the outcome variable.

- Wrapper Methods

- Uses a **specific model's performance to evaluate subsets of features**.
- Based on the inferences that we draw from the previous model, we decide to add or remove features from your subset.
- These methods are usually computationally very expensive

- Embedded Methods

- **Incorporate feature selection within the model training process.**

Correlation Feature Selection (CFS):

Correlation-based feature ranking: A made-up dataset

Which features are related to the class?

f1	f2	f3	f4	...	class
0.4	0.6	0.4	0.6		1
0.2	0.4	1.6	-0.6		1
0.5	0.7	1.8	-0.8		1
0.7	0.8	0.2	0.9		2
0.9	0.8	1.8	-0.7		2
0.5	0.5	0.6	0.5		2

Correlated with the class

f1	f2	f3	f4	...	class
0.4	0.6	0.4	0.6		1
0.2	0.4	1.6	-0.6		1
0.5	0.7	1.8	-0.8		1
0.7	0.8	0.2	0.9		2
0.9	0.8	1.8	-0.7		2
0.5	0.5	0.6	0.5		2

Uncorrelated with the class / seemingly random

f1	f2	f3	f4	...	class
0.4	0.6	0.4	0.6		1
0.2	0.4	1.6	-0.6		1
0.5	0.7	1.8	-0.8		1
0.7	0.8	0.2	0.9		2
0.9	0.8	1.8	-0.7		2
0.5	0.5	0.6	0.5		2

Correlation based FS reduces the dataset to this

f1	f2			...	class
0.4	0.6				1
0.2	0.4				1
0.5	0.7				1
0.7	0.8				2
0.9	0.8				2
0.5	0.5				2

Problem with correlation-based FS (CFS):

Column 5 shows us $f3 + f4$ – which is perfectly correlated with the class!

f1	f2	f3	f4	...	class
0.4	0.6	0.4	0.6	1	1
0.2	0.4	1.6	-0.6	1	1
0.5	0.7	1.8	-0.8	1	1
0.7	0.8	0.2	0.9	1.1	2
0.9	0.8	1.8	-0.7	1.1	2
0.5	0.5	0.6	0.5	1.1	2

Feature Selection: Challenges

- **Curse of Dimensionality:**

In high-dimensional spaces, finding the optimal subset of features becomes computationally challenging.

- **Interaction Effects:**

Feature selection might overlook interactions between features, which can be important in some cases.

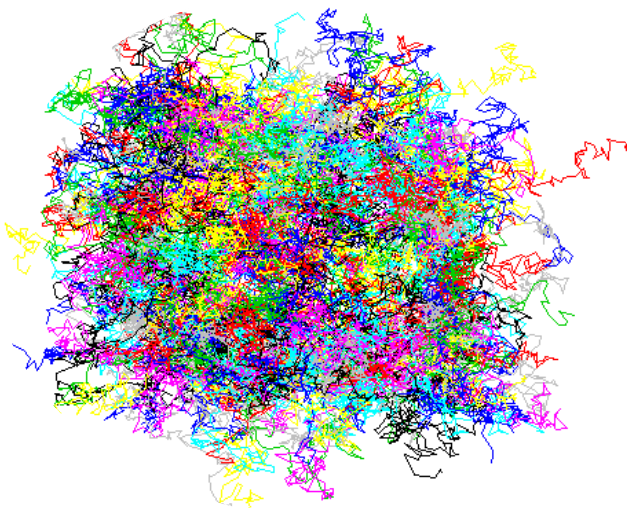
- **Dynamic Data:**

Feature importance can change over time, requiring periodic re-evaluation of feature selection.

Dimensionality Reduction

Why reduce dimensionality in my data?

- ❑ Simplifies Data Structure
- ❑ Optimizes Performance in Machine Learning.
- ❑ Identify patterns in data
- ❑ Visualise high-dimensional data



WIZARD_Title

Data Table

Point Of View

Display

Worksheet POV

4 Entity members

4 Period members

Actual

2020

<Entity Curr Total>

<Scenario View>

Net Assets

[ICP Top]

Total

Total

Total

Total

Not Used

[None]

Scenario	Year	Period	View	Entity	Account	ICP	Custom1	Custom2	Custom3	Custom4	Custom5	Value	V&B Entity	(Ex Rate)	V&B GP Tot
Actual	2020	MAY	YTD	PLC	110110	[ICP Top]	TopC1	CPX	TopC3	TopC4	[None]	[None]	GBP Total	(838)	1.0000 (838)
Actual	2020	APR	YTD	NAMTOT	227000	[ICP Top]	TopC1	OPENBAL	TopC3	TopC4	[None]	[None]	GBP Total	(470)	1.0000 (470)
Actual	2020	MAY	YTD	PLC	10	[ICP Top]	TopC1	DIV7	TopC3	TopC4	[None]	[None]	GBP Total	(554)	1.0000 (554)
Actual	2020	APR	YTD	EURTOT	11	[ICP Top]	TopC1	GRTTOT	TopC3	TopC4	[None]	[None]	GBP Total	101	1.0000 101
Actual	2020	MAY	YTD	PLC	11001	[ICP Top]	TopC1	CPX	TopC3	TopC4	[None]	[None]	GBP Total	20,361	1.0000 20,361
Actual	2020	MAY	YTD	PLC	229500	[ICP Top]	TopC1	DIV5	TopC3	TopC4	[None]	[None]	GBP Total	46	1.0000 46
Actual	2020	MAY	YTD	EURTOT	224000	[ICP Top]	TopC1	BST	TopC3	TopC4	[None]	[None]	GBP Total	(883)	1.0000 (883)
Actual	2020	MAY	YTD	NAMTOT	PENPROV	[ICP Top]	TopC1	PRV	TopC3	TopC4	[None]	[None]	GBP Total	(141,030)	1.0000 (141,030)
Actual	2020	MAY	YTD	EURTOT	PLANT	[ICP Top]	TopC1	LYACQ7	TopC3	TopC4	[None]	[None]	GBP Total	(2)	1.0000 (2)
Actual	2020	JUL	YTD	PLC	TRADEPAYSANDBILLS	[ICP Top]	TopC1	DIV10	TopC3	TopC4	[None]	[None]	GBP Total	2,089	1.0000 2,089
Actual	2020	JUL	YTD	EURTOT	OPLEASELB	[ICP Top]	TopC1	LYACQ7	TopC3	TopC4	[None]	[None]	GBP Total	(1)	1.0000 (1)
Actual	2020	MAY	YTD	PLC	229500	[ICP Top]	TopC1	EXD	TopC3	TopC4	[None]	[None]	GBP Total	(575)	1.0000 (575)
Actual	2020	MAY	YTD	PLC	211000	[ICP Top]	TopC1	WOF	TopC3	TopC4	[None]	[None]	GBP Total	56,394	1.0000 56,394
Actual	2020	JUL	YTD	UKITOT	200000	[ICP Top]	TopC1	DIV1	TopC3	TopC4	[None]	[None]	GBP Total	(44,683)	1.0000 (44,683)
Actual	2020	MAY	YTD	PLC	VEHICLES	[ICP Top]	TopC1	DIS	TopC3	TopC4	[None]	[None]	GBP Total	(6,516)	1.0000 (6,516)
Actual	2020	MAY	YTD	EURTOT	OWNEDBL	[ICP Top]	TopC1	EXD	TopC3	TopC4	[None]	[None]	GBP Total	49,301	1.0000 49,301
Actual	2020	JUN	YTD	EURTOT	120010	[ICP Top]	TopC1	OPENBAL	TopC3	TopC4	[None]	[None]	GBP Total	39	1.0000 39
Actual	2020	MAY	YTD	PLC	FINLEASELB	[ICP Top]	TopC1	DIS	TopC3	TopC4	[None]	[None]	GBP Total	(4,452)	1.0000 (4,452)
Actual	2020	JUL	YTD	PLC	101030	[ICP Top]	TopC1	OPENBAL	TopC3	TopC4	[None]	[None]	GBP Total	(1,739)	1.0000 (1,739)
Actual	2020	JUN	YTD	EURTOT	21	[ICP Top]	TopC1	DIV6	TopC3	TopC4	[None]	[None]	GBP Total	(1,530)	1.0000 (1,530)
Actual	2020	MAY	YTD	PLC	110020	[ICP Top]	TopC1	WIP	TopC3	TopC4	[None]	[None]	GBP Total	(22,209)	1.0000 (22,209)
Actual	2020	JUL	YTD	PLC	101050	[ICP Top]	TopC1	OPENBAL	TopC3	TopC4	[None]	[None]	GBP Total	7	1.0000 7
Actual	2020	MAY	YTD	NAMTOT	101	[ICP Top]	TopC1	CPX	TopC3	TopC4	[None]	[None]	GBP Total	1,945	1.0000 1,945
Actual	2020	JUL	YTD	PLC	10100	[ICP Top]	TopC1	DIV7	TopC3	TopC4	[None]	[None]	GBP Total	(279)	1.0000 (279)
Actual	2020	JUN	YTD	PLC	202000	[ICP Top]	TopC1	DIV1	TopC3	TopC4	[None]	[None]	GBP Total	4,880	1.0000 4,880
Actual	2020	APR	YTD	EURTOT	12	[ICP Top]	TopC1	PRV	TopC3	TopC4	[None]	[None]	GBP Total	(13)	1.0000 (13)
Actual	2020	JUN	YTD	PLC	FINLEASELB	[ICP Top]	TopC1	DIS	TopC3	TopC4	[None]	[None]	GBP Total	(4,489)	1.0000 (4,489)
Actual	2020	MAY	YTD	PLC	110300	[ICP Top]	TopC1	CAT	TopC3	TopC4	[None]	[None]	GBP Total	102	1.0000 102
Actual	2020	JUN	YTD	EURTOT	110005	[ICP Top]	TopC1	DIV5	TopC3	TopC4	[None]	[None]	GBP Total	45	1.0000 45
Actual	2020	JUL	YTD	PLC	10	[ICP Top]	TopC1	DIV7	TopC3	TopC4	[None]	[None]	GBP Total	(556)	1.0000 (556)
Actual	2020	JUL	YTD	PLC	21	[ICP Top]	TopC1	ABSVAR	TopC3	TopC4	[None]	[None]	GBP Total	884,149	1.0000 884,149
Actual	2020	JUN	YTD	PLC	228000	[ICP Top]	TopC1	EXD	TopC3	TopC4	[None]	[None]	GBP Total	(8,534)	1.0000 (8,534)
Actual	2020	APR	YTD	PLC	20	[ICP Top]	TopC1	SET	TopC3	TopC4	[None]	[None]	GBP Total	(175,082)	1.0000 (175,082)
Actual	2020	MAY	YTD	PLC	229	[ICP Top]	TopC1	DIV5	TopC3	TopC4	[None]	[None]	GBP Total	26	1.0000 26
Actual	2020	JUN	YTD	UKITOT	110200	[ICP Top]	TopC1	CPX	TopC3	TopC4	[None]	[None]	GBP Total	20,445	1.0000 20,445

Last refreshed: 12 May 2021 09:37 17,503 rows. Refresh took 21 seconds

Display Settings

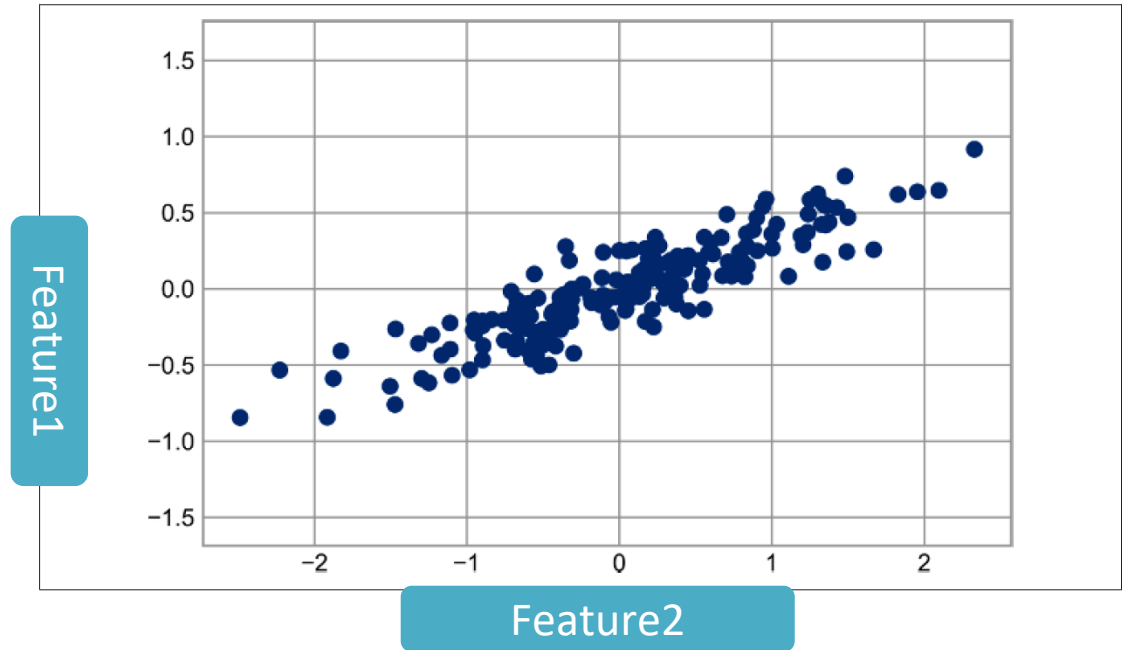
Principal Component Analysis

- **Principal Component Analysis (PCA)** is a **dimensionality reduction** technique commonly used in data analysis and machine learning.
- It is a powerful technique that arises from linear algebra and probability theory. In essence, **it is an algorithm that computes a matrix that represents the variation of your data** (covariance matrix/eigenvectors), **and rank the vectors of the matrix by their relevance.**
- It transforms a large set of possibly **correlated variables** into a smaller set of **uncorrelated variables** called **principal components**, while **retaining as much variation (information) as possible.**

Why use PCA

By the eye, are these two features correlated?

Do we want correlated features in training our model?



PCA decorrelates these two features by finding the Principal Component.

Why we want to decorrelate two features?

1. Simplifies Data Structure

- ❑ In datasets with high correlations between features, some features carry **redundant information** (e.g. if height and weight are highly correlated, they give redundant information if I am predicting diabetes).

2. Compression and faster training

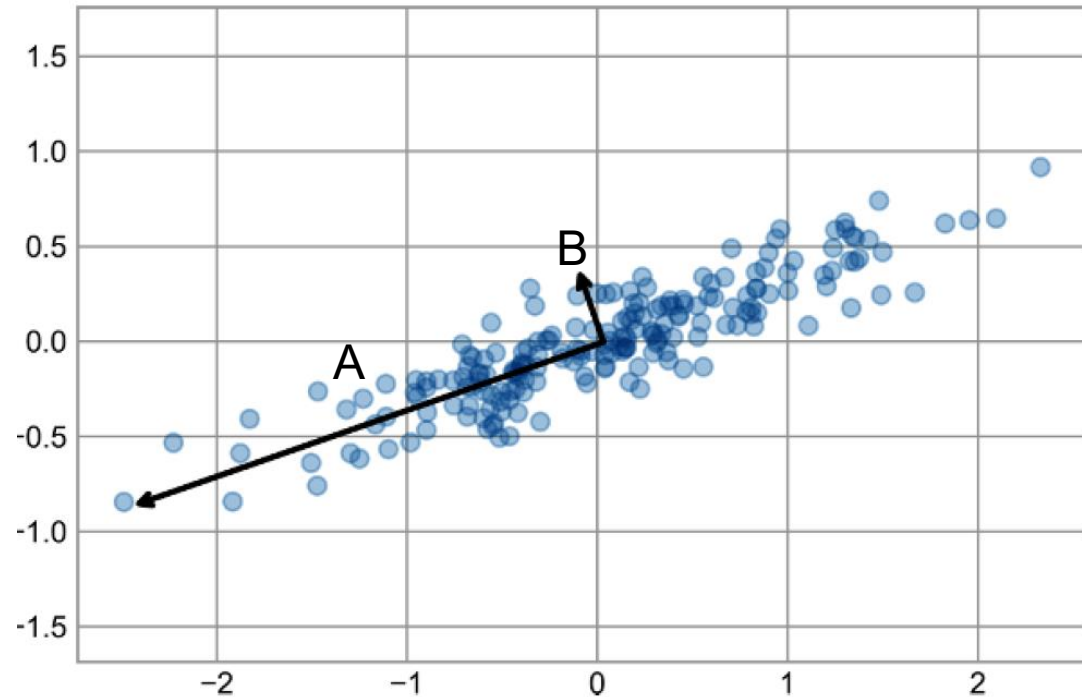
- ❑ Reducing the number of features with PCA makes training much faster and better for visualization purposes.

3. Optimizes Performance in Machine Learning

- ❑ Many machine learning algorithms (e.g., linear regression, k-means clustering) assume or perform better when input features are **independent**.

What are the principal components in PCA?

- The PCA algorithms finds the relationship between the features and quantifies this relationship by finding a list of the **principal axes** in the data.
- The principal axes are vectors that describe the distribution of the data.
- The length of each vector is an indication of how “important” that axis in describing the distribution of the data.
- Which one is more important (A or B) in the figure?
- The projection of each data point onto the principal axes are the **principal components** of the data

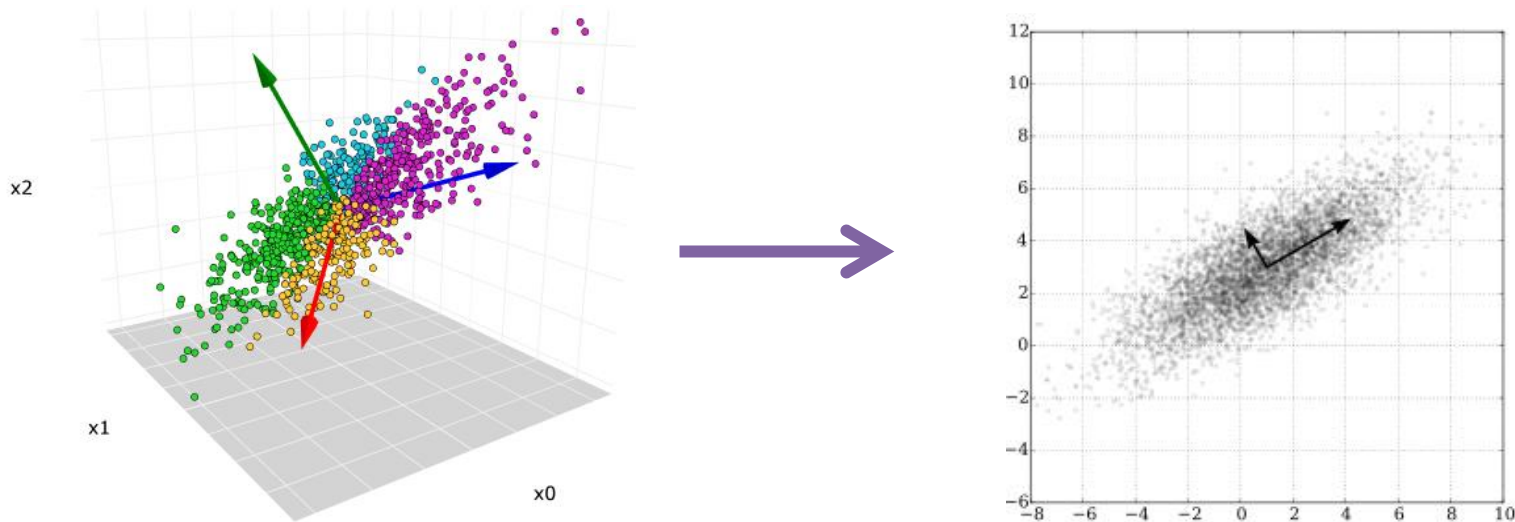


Visualization of the principal axes in the data

Applying PCA to the data

*Remember: Using PCA for dimensionality reduction involves **zeroing out** one or more of the smallest principal components, resulting in a lower-dimensional projection of the data that preserves the maximal data variance (maximum information).*

This may lead to information loss but it can also sometimes lead to better performance (faster and simpler model)



Visualisation of how PCA works

[Principal Component Analysis explained visually](#)

[Colab](#)

When to Use PCA?

- When your dataset has **many features**, and you suspect some of them are redundant or highly correlated.
- To **visualize high-dimensional data** in 2D or 3D.
- As a **preprocessing step** before applying machine learning algorithms to improve performance.

Pros and Cons:

Pros:

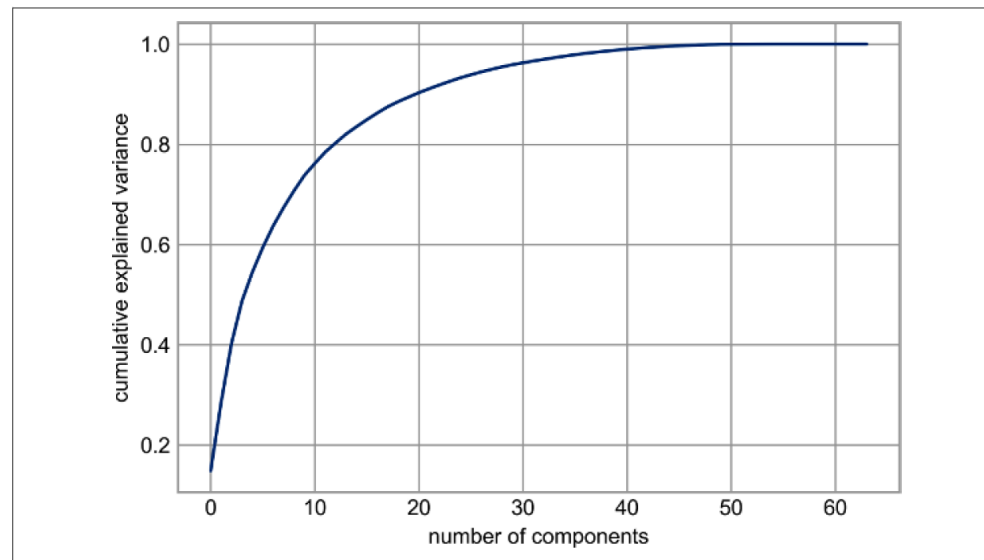
- ❖ Reduces computational complexity.
- ❖ Helps avoid overfitting.
- ❖ Makes data visualization easier.

Cons:

- Can be hard to interpret the new principal components.
- Some information is inevitably lost during dimensionality reduction.
- Assumes linear relationships between variables.
- **PCA's main weakness is that it tends to be highly affected by outliers in the data.**

Choosing the Number of Components in PCA

- A vital part of using PCA in practice is the ability to estimate how many components are needed to describe the data.
- This can be determined by looking at the *cumulative explained variance ratio* as a function of the number of components.



- This curve quantifies how much of the total information is contained within the first N components.
- In this figure, we'd need about 20 components to retain 90% of the variance (information in the data). We will learn how to calculate this in the lab session.

Feature preprocessing: Outliers

What is an outlier?

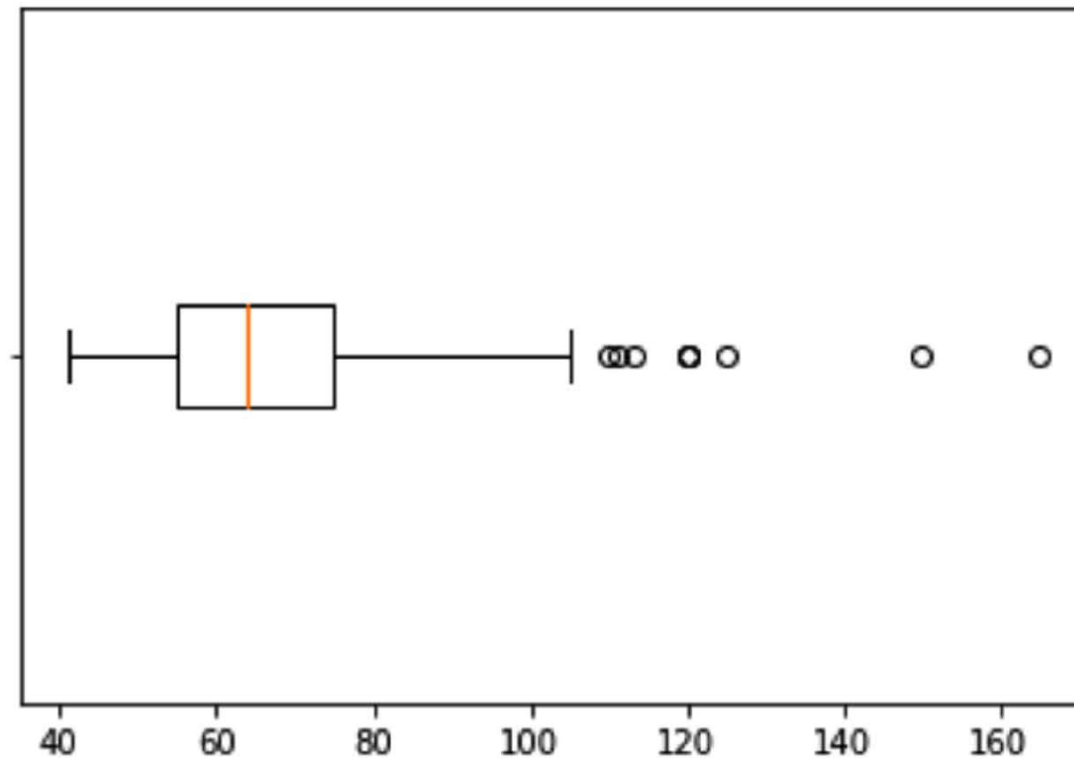
Outliers, a.k.a. extreme points, are data objects whose values are too different than the rest of the population.

How can outliers affect your machine learning model?

How can we detect outliers?

Univariate outlier detection

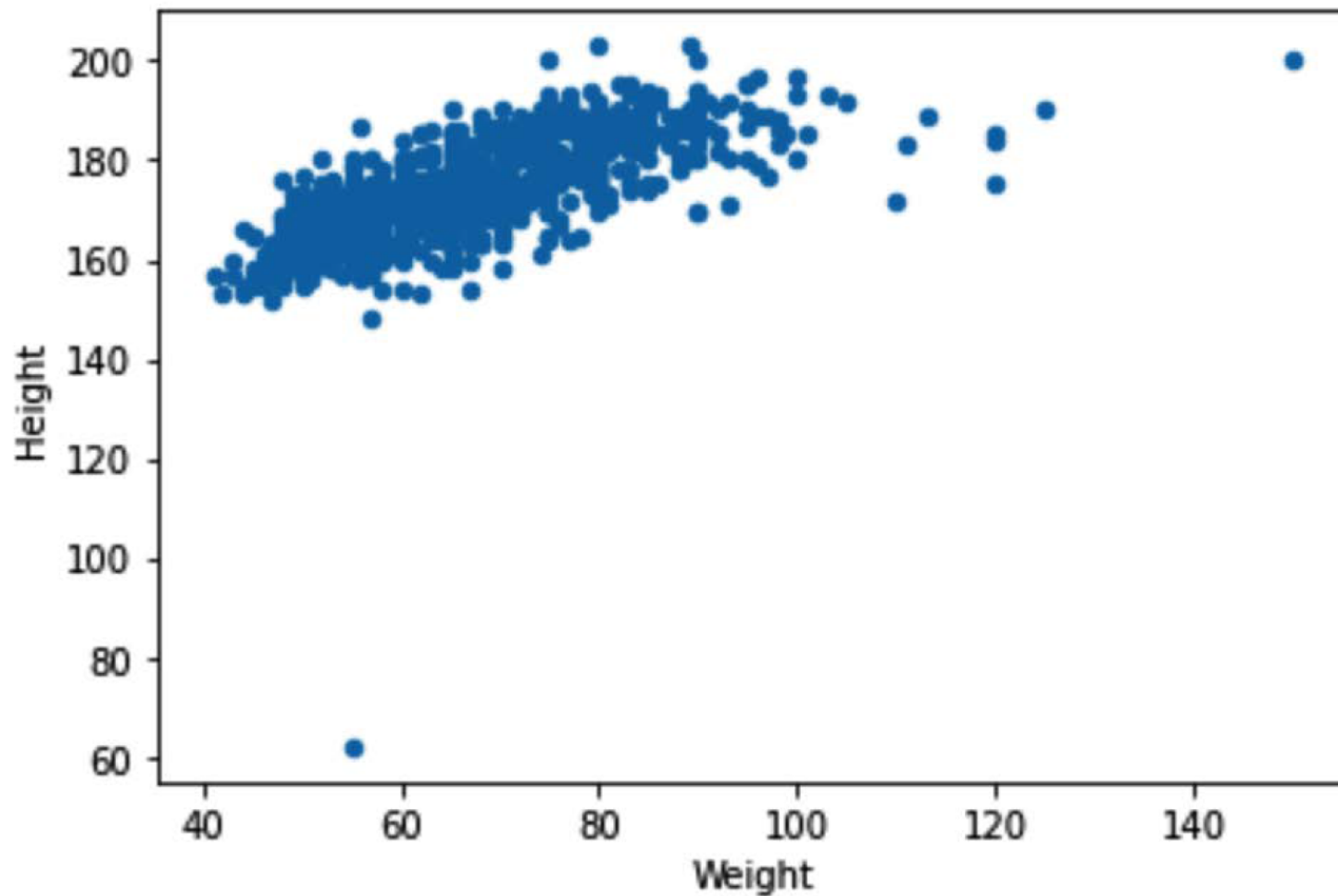
A boxplot if it is a **numerical** data and the small circles (fliers) represent the outliers for this single feature (e.g weight)



For detecting outliers across one **categorical** attribute, we can use a frequency table or a bar chart.

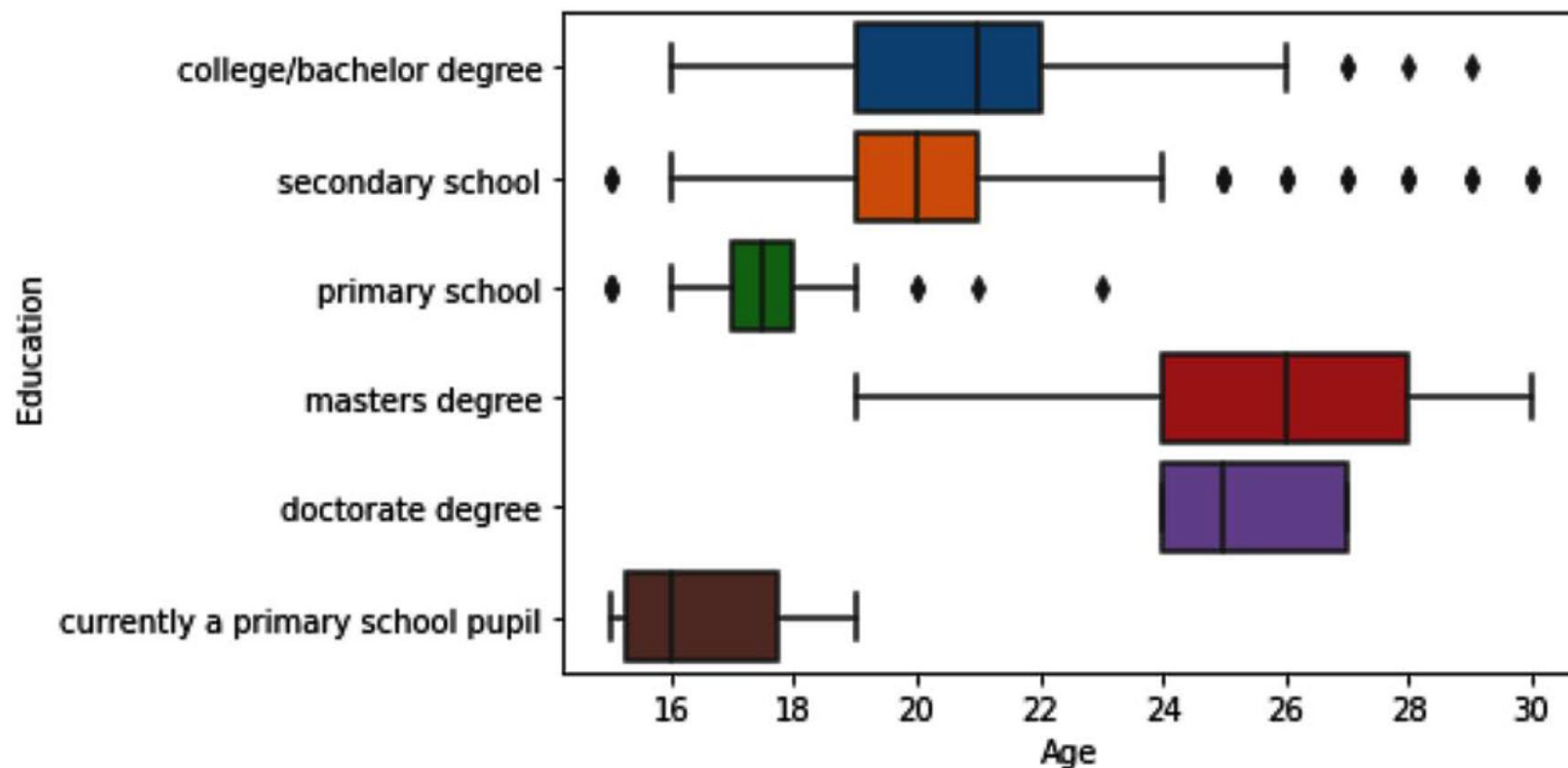
Bivariate outlier detection

What are the outliers in this graph?



Bivariate outlier detection

Here the two variables are different: one is categorical, and the other is numeric.

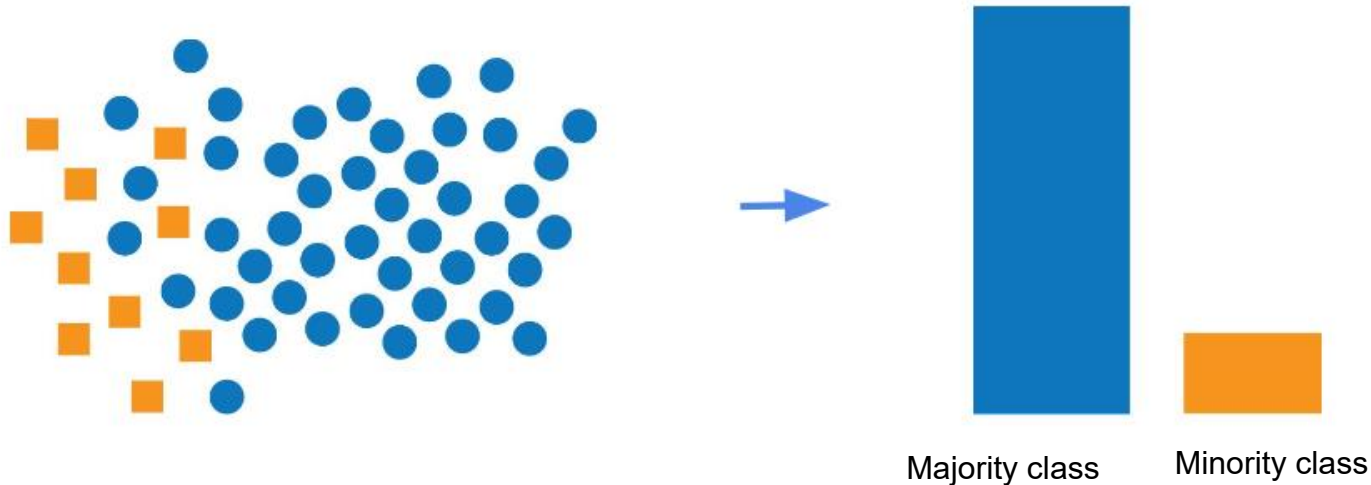


How to deal with outliers?

- Do nothing
 - Replace with the upper cap or lower cap
 - Perform a log transformation
 - Remove data objects with outliers
-
- ❑ Whether to remove or retain outliers depends on the specific problem:
 - ❑ Evaluate how handling outliers affects your model's performance.
 - ❑ In some cases, domain expertise can clarify if outliers are noise or valuable signals.

Unbalanced Data: what to do?

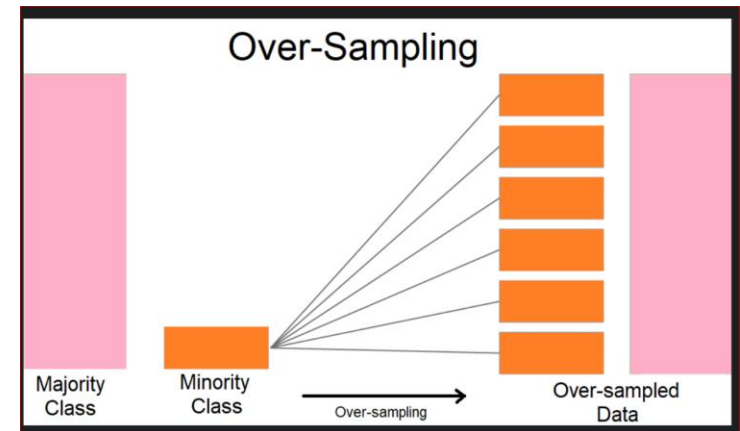
Unbalanced data is when the dataset has significantly more samples from one class than another (e.g., fraud detection, medical diagnosis).



Solutions for unbalanced data

1. Resampling Techniques

a. Oversampling the Minority Class

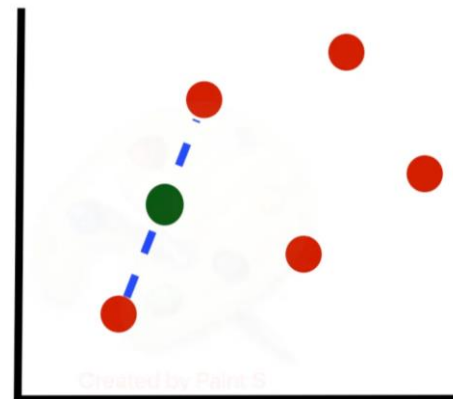


➤ Duplicate or synthesize new samples for the minority class.

➤ **Methods:**

1. **Random Oversampling:** **Randomly** duplicate samples from the minority class.
2. **SMOTE (Synthetic Minority Oversampling Technique):**
 1. Take difference between a sample and its nearest neighbour.
 2. Multiply the difference by a random number between 0 and 1
 3. Add this difference to the sample to generate a new synthetic example in feature space.

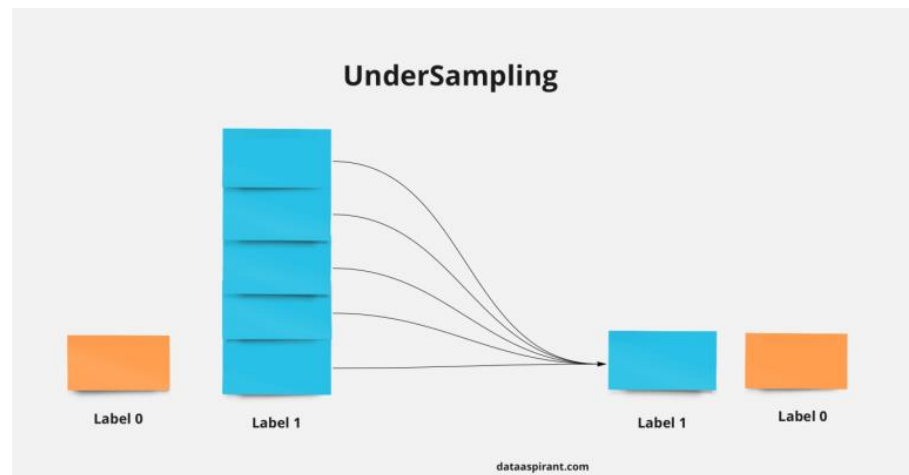
We compute the difference between the two points and multiply it by a random number between 0 and 1. This gives us a synthetic example (the green circle) along the line between the two points.



2. Resampling Technique

b. Undersampling the Majority Class

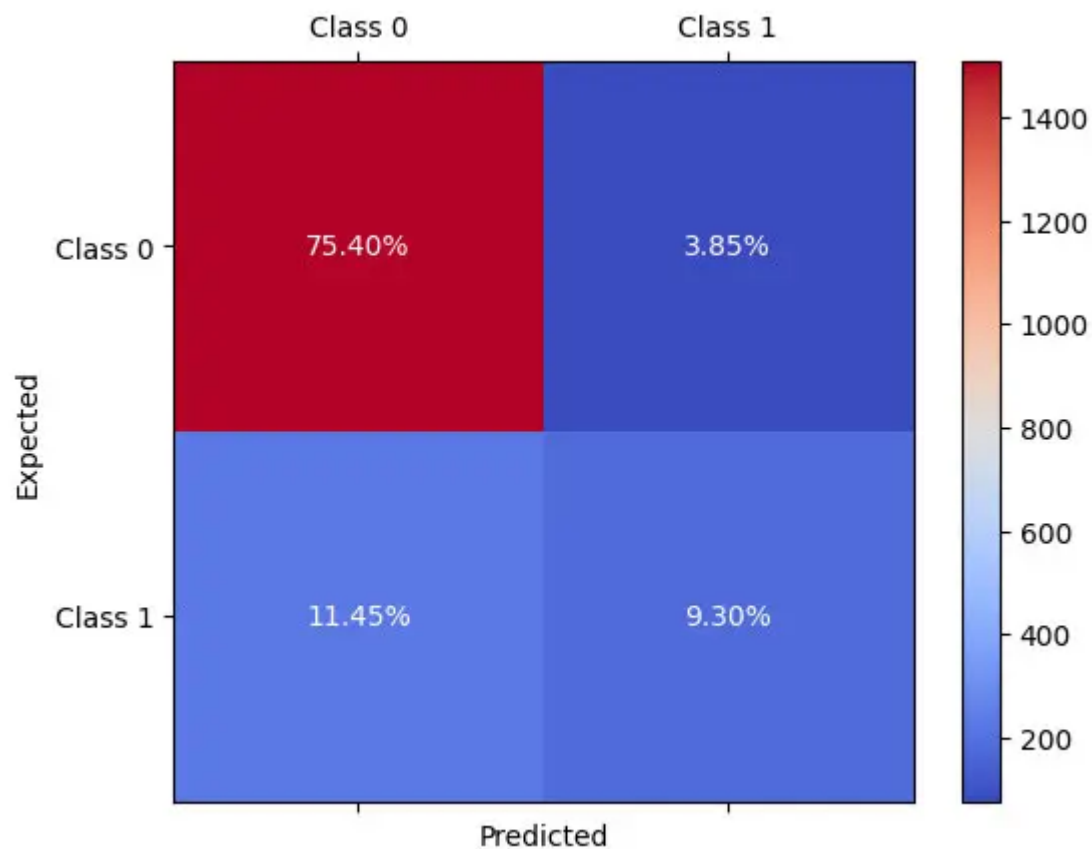
- **Technique:** Reduce the number of samples from the majority class.
- **Methods:**
 1. **Random Undersampling:** Randomly remove samples from the majority class.
 2. **NearMiss:** Selects samples closest to the minority class and deletes them.



Evaluation Metrics for Imbalanced Data

- ❑ Accuracy is often misleading with imbalanced datasets.
- ❑ Use metrics that reflect the performance on both classes:
- ❑ **Precision**: How many predicted positives are actual positives.
- ❑ **Recall (Sensitivity)**: How many actual positives are correctly predicted.
- ❑ **F1-Score**: Harmonic mean of Precision and Recall.
- ❑ **ROC-AUC**: Evaluates the model's ability to distinguish between classes.

How do you interpret this confusion metric?



Assessment 1

How far are you in finishing your experiment?

What are the problems you have so far?

What we learned today

1. Dimensionality Reduction
2. Principal Component Analysis
3. Detecting and dealing with outliers
4. Dealing with unbalanced data.
5. Checking on assessment 1